

AIM 240: AI/ML Capstone Project Initialization Document

Student Name:

Semester/Year: Spring 2026

Introduction

Welcome to your Capstone Project! This notebook will guide you through the process of defining, scoping, and planning your individual AI/ML project. Since this course is asynchronous, this document serves as our initial communication tool for understanding your vision and providing targeted feedback.

Please complete ALL sections thoroughly. Incomplete submissions will be returned for revision before approval.

How to Use This Notebook

1. Read through the entire notebook first to understand what's expected
 2. Complete Section 1 (Project Ideation) with 2-3 potential project ideas
 3. Use Section 2 to systematically evaluate and rank your ideas
 4. Complete Sections 3-12 for your **top-ranked project only**
 5. Export as HTML or PDF and submit via the course LMS by the posted deadline
 6. Await instructor feedback before beginning implementation
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Section 1: Project Ideation

1.1 Brainstorming Your Project Ideas

Project A

Working Title: [Generative Diffusion Model for MHD/PIC Plasma Physics Surrogate]

Brief Description:

[Using a generative diffusion model, borrowing architecture from foundational 2020 DDPM paper, to predict plasma dynamics and turbulence. Diffusion model will be adapted to adhere to physical constraints to probabilistically capture plasma density progression, velocity field changes, and magnetic field evolution. A diffusion model will make plasma physics simulations significantly more efficient and faster, enabling advancement in regimes like nuclear fusion and active control of fusion technologies.]

Primary ML Domain(s): [Select all that apply: Computer Vision, Tabular/Structured Data, Time Series/Forecasting, Generative AI, Multi-modal]

Project B

Working Title: [Fourier Neural Operators for Weather Data and Natural Disaster Prediction]

Brief Description:

[Using Fourier Neural Operators to capture chaotic weather dynamics by emulating fluid-flow PDEs with neural operator representations. Will predict the evolution of atmospheric pressure, temperature, and windspeed. Extension of FourCastNet model by NVIDIA to investigate long term climate dynamics for the purposes of anticipating serious climate change events.]

Primary ML Domain(s): [, Tabular/Structured Data, Time Series/Forecasting, Neural Operators]

✓ Section 2: Project Ranking & Selection

2.1 Evaluation Criteria Matrix

Rate each of your project ideas on a scale of **1-5** for each criterion below. Be honest in your assessment - this exercise helps you avoid projects that may become problematic later.

Rating Scale:

- 1 = Very Poor / Major Concerns
 - 2 = Below Average / Some Concerns
 - 3 = Adequate / Neutral
 - 4 = Good / Minor Concerns
 - 5 = Excellent / No Concerns
-

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# =====
# PROJECT RANKING CALCULATOR
# Fill in your scores below (1-5 for each criterion)
# =====

import pandas as pd

# Define the evaluation criteria
criteria = [
    "Personal Interest",      # How excited are you about this project?
    "Data Availability",      # Is appropriate data publicly available?
    "Technical Feasibility",  # Can this be accomplished with your current skill?
    "Scope Appropriateness", # Is this achievable in one semester?
    "Novelty/Learning Value", # Will you learn new skills beyond what you know?
    "Portfolio Value",        # Will this impress future employers?
    "Ethical Clarity",        # Are there minimal ethical concerns?
    "MLOps Applicability"     # Can you apply MLOps concepts?
]

# =====
# ENTER YOUR SCORES HERE (1-5 for each)
# =====

# Project Idea A scores
idea_a_scores = {
    "Personal Interest": 5,      # <-- Enter score 1-5
    "Data Availability": 3,      # <-- Enter score 1-5
    "Technical Feasibility": 3,  # <-- Enter score 1-5
    "Scope Appropriateness": 2, # <-- Enter score 1-5
    "Novelty/Learning Value": 4, # <-- Enter score 1-5
    "Portfolio Value": 5,        # <-- Enter score 1-5
    "Ethical Clarity": 5,        # <-- Enter score 1-5
    "MLOps Applicability": 3     # <-- Enter score 1-5
}

# Project Idea B scores
idea_b_scores = {
    "Personal Interest": 4,      # <-- Enter score 1-5
    "Data Availability": 3,      # <-- Enter score 1-5
    "Technical Feasibility": 3,  # <-- Enter score 1-5
    "Scope Appropriateness": 2, # <-- Enter score 1-5
    "Novelty/Learning Value": 3, # <-- Enter score 1-5
    "Portfolio Value": 3,        # <-- Enter score 1-5
    "Ethical Clarity": 5,        # <-- Enter score 1-5
    "MLOps Applicability": 3     # <-- Enter score 1-5
}

# Project Idea C scores (set to 0 if not using)
idea_c_scores = {
    "Personal Interest": 0,      # <-- Enter score 1-5 (or leave as 0)
    "Data Availability": 0,      # <-- Enter score 1-5 (or leave as 0)

```

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    "Technical Feasibility": 0,      # <-- Enter score 1-5 (or leave as 0)
    "Scope Appropriateness": 0,    # <-- Enter score 1-5 (or leave as 0)
    "Novelty/Learning Value": 0,   # <-- Enter score 1-5 (or leave as 0)
    "Portfolio Value": 0,          # <-- Enter score 1-5 (or leave as 0)
    "Ethical Clarity": 0,         # <-- Enter score 1-5 (or leave as 0)
    "MLOps Applicability": 0      # <-- Enter score 1-5 (or leave as 0)
}

df_scores = pd.DataFrame({
    "Criterion": criteria,
    "Idea A": [idea_a_scores[c] for c in criteria],
    "Idea B": [idea_b_scores[c] for c in criteria],
    "Idea C": [idea_c_scores[c] for c in criteria]
})

totals = pd.DataFrame({
    "Criterion": ["TOTAL SCORE"],
    "Idea A": [sum(idea_a_scores.values())],
    "Idea B": [sum(idea_b_scores.values())],
    "Idea C": [sum(idea_c_scores.values())]
})

df_display = pd.concat([df_scores, totals], ignore_index=True)

print("=" * 60)
print("PROJECT EVALUATION MATRIX")
print("=" * 60)
print(df_display.to_string(index=False))
print("\n" + "=" * 60)
print(f"Maximum possible score: 40")
print("=" * 60)

```

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PROJECT EVALUATION MATRIX
=====

```

Criterion	Idea A	Idea B	Idea C
Personal Interest	5	4	0
Data Availability	3	3	0
Technical Feasibility	3	3	0
Scope Appropriateness	2	2	0
Novelty/Learning Value	4	3	0
Portfolio Value	5	3	0
Ethical Clarity	5	5	0
MLOps Applicability	3	3	0
TOTAL SCORE	30	26	0

```

=====
Maximum possible score: 40
=====

```

2.2 Final Ranking & Justification

Final Ranking

1. Generative Diffusion Model for Plasma Physics: This is my top choice as it is most relevant to my field of study, fusion sciences. ML is becoming a big part of fusion research due to the complexity and tenacity of the physics involved, and there is sparse research on how newer models can be utilized. There are not many existing frameworks for me to take inspiration from, so if this turns out to be too difficult then I will switch to 2nd choice.
 2. Fourier Neural Operator for Weather and Natural Disaster Prediction: This project is also of interest to me to the success of neural operators in serving as PDE surrogates for chaotic dynamics. This is particularly useful for weather models as current models consist of sophisticated nested PDEs with grid points across the globe. FNOs have already been used successfully for weather modeling and I want to apply it to specific subjects in the field of weather science/climate prediction.
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✓ Section 3: Problem Statement

Complete this section and all following sections for your TOP-RANKED PROJECT ONLY.

Project A

3.1 Problem Statement

Magnetohydrodynamic (MHD)/particle-in-cell (PIC) simulations are the mechanisms to study plasma dynamics relevant fusion science and astrophysics. They rely on solving the governing electrodynamics and fluid flow PDE's on the nanosecond timescale. These multi-physics simulations are very computationally and labor intensive, there has been research to use statistical and ML surrogates. I plan to investigate the use of diffusion models as it they present the opportunity to capture multi-modal datasets and uncover novel physical representations.

3.2 Existing Solutions

Solution: HPC and technological advancement, Type: Hardware, Strengths: Allows for full range of physics to be covered with more efficient compute, Weaknesses: Relies on industry advancement and technically difficult to realize. Why mine is different: Addresses problem from software standpoint while focusing on the mai

Solution: Other AI/ML surrogate models, Type: AI/ML, Strengths: Has demonstrated success both from fundamental plasma physics perspective and in application with places like NIF, Weaknesses: ML paradigms are advancing fast and there is sparse research on how new models can be utilized. Some models like the ability to be multi-modal as well

Project B

3.1 Problem Statement

Weather patterns are notoriously difficult to predict due to their chaotic nature and the multitude of variables it is sensitive to. Developing more efficient, accurate models is of grave interest to humanity due to the impending threats of climate change. I want to use a Fourier Neural Operator framework to predict long term climate changes based on historical ERA5 data, expanding on research done with NVIDIA's FourCastNet.

3.2 Existing Solutions

Solution: FourCastNet by NVIDIA, Type: ML/AI, Strengths: More effective than other ML modalities, offers much faster inference and better results than trad models, Weaknesses: Focused on short term casting, Where mine is different: Extending the model (either by fine tuning or by direct manipulation) to focus on climate change and long term dynamics

Solution: Traditional numerical models, Type: Mathematical modeling, Strengths: Physically consistent, most accurate in short/long time scale, Weaknesses: Incredibly computationally expensive, limited spatial resolution, physics is estimated. Where mine is different: Finds representation of the PDEs involved in order to have similar accuracy, fast inference, can be optimized to capture multi-scale physics

✓ Section 4: Project Scope Definition

Project A

4.1 Essential

1. Feature: Regression, Descr: Predict scalar, 2D density fluctuation, Success: 80% accuracy
2. Feature: Classification, Descr: Classify the type of instabilities occurring, Success: >80% Accuracy

3. Feature: Physics consistency, Descr: Model constrained by physics, model parameters obey conservation laws, Success: comparable to PDE models

4.2 Nice to Have

1. Feature: Interactive UI, Descr: UI where users are able to manipulate input parameters in real time and explore the physical model, Success: Other classmates are able to use UI. Dependencies: Primary objectives are met and time allows
2. Feature: Compact model, Descr: Model uses as little parameters/nodes as possible, efficient, Success: Inference time <10s, Dependencies: Model excels at primary features 1 and 3
3. Feature: Conditional generation, Descr: Model is not just dependent on the initial state, but user chosen control parameters like turbulence intensity (reynolds number), Success: Comparable to PDE models, Dependencies: Model excels at primary features 1 and 3

4.3 Out of scope

1. Connection to ICF experiment and data (i.e. NIF)
2. Structure for real time control of tokamak fusion modalities
3. Inclusion of laser or microwave driving

4.4 Scope Validation Checklist

Answer honestly by changing ☐ to ☒ for items that apply:

- ☒ Can I complete the core features in 10-12 weeks of part-time work?
- ☒ Have I built something of similar complexity before, OR have I allocated time to learn?
- ☒ If I removed all extended features, would I still have a meaningful project?
- ☒ Is this project substantially different from a homework assignment or tutorial?
- ☐ Does this project require me to stretch my skills without being overwhelming?

If you checked fewer than 4 boxes, reconsider your scope.

Project B

4.1 Essential Features

1. Feature: Atmospheric Pressure Distribution, Descr: Predict how atmospheric pressure changes regionally day to day, Success: >60% Accuracy
2. Feature: Macroscale global temperature anomalies, Descr: Predict sharp deviations in climate that are significant, Success: Comparison to historical data

3. Feature: UI and visualization, Descr: User friendly interface that allows one to set initial conditions and watch how weather progresses with 2D graphic, Success: Accessible to 3 classmates

4.2 Nice to have

1. Feature: Real time weather feedback, Descr: Able to scrape weather data and make real time predictions, Dependencies: Excellent accuracy and fidelity in trained model, >80%
2. Feature: Compact model, Descr: Fewer nodes and compartments in model, able to find the most effective representation of physics involved, Dependencies: Physics validation of initial model

4.3 Out of scope

1. High fidelity, hour to hour, weather predictions
2. City level, microscale predictions/high spatial resolution

4.4 Scope Validation Checklist

Answer honestly by changing ☐ to ☒ for items that apply:

- ☒ Can I complete the core features in 10-12 weeks of part-time work?
 - ☒ Have I built something of similar complexity before, OR have I allocated time to learn?
 - ☒ If I removed all extended features, would I still have a meaningful project?
 - ☒ Is this project substantially different from a homework assignment or tutorial?
 - ☐ Does this project require me to stretch my skills without being overwhelming?
-

✓ Section 5: Data Strategy

Project A

5.1 Data Requirements

1. Datatype: Plasma Field Data, Description: MHD simulation data of plasma density, velocity field of plasma, magnetic field density/field dynamics, Format: Scalar and vector data, CSV

5.2 Data Sources

1. Numerical Sim Data

- Name: John Hopkins Turbulence Database; <https://turbulence.idies.jhu.edu/home>
Type: Public Access: Free download License: Open
access; <http://opendatacommons.org/licenses/by/> Quality Assessment: Well known
source, JHU and Los Alamos, physically based model Volume: 50 TB

2. Curated Sim Data

- Name: MHD Solver, Dedalus Project; <https://github.com/DedalusProject/dedalus>
Type: Public Access: Free download from Git License: MIT License Quality
Assessment: Developed by MIT researchers Volume: Arbitrary, self-generated

5.3 Data Collection Plan

The JHU database is readily available for download as a CSV. Upon confirmation of final project choice, I will download this dataset, and take 2 weeks to format data according to model specifications and needs. During this data formatting process, I will use Dedalus to generate my own MHD data in a way that diversifies the dataset and is curated to the needs the JHU database can't fulfill

5.4 Data Validation Checklist

Answer honestly by changing ☐ to ☒ for items that apply:

- ☒ I have verified that my primary data source is accessible right now
- ☒ I have reviewed the license and confirmed I can use this data for my project
- ☐ I have examined sample data and understand its structure
- ☒ I have a backup plan if my primary data source becomes unavailable
- ☒ My data volume is sufficient for training a meaningful model
- ☐ I have considered and documented any data quality issues

Backup Data Plan:

I will utilize MHD solvers to generate the desired data

Project B

5.1 Data Requirements

1. Datatype: Temperature, Pressure, Wind, Descr: Scalar data of temp, air pressure, and horizontal wind at each grid point around globe. Format: Scalar, CSV data

5.2 Data Sources

- Name: ERA5, <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=download> Type: Public access, EU/government database License: CC-

BY licence Quality assessment: Internationally recognized, used in other successful weather models

- Name: Weatherbench, <https://github.com/pangeo-data/WeatherBench/blob/master/README.md> Type: Git, public access License: MIT License Quality assessment: Inherits from ERA5, preprocesses data in ML interpretable format. Reliable

5.3 Data collection plan

I will explore ERA5 and Weatherbench to see how the datasources are formatted against each other. If this is the project I opt for, then I will spend 3 weeks parsing through the data and curating it to my needs. I want to predict long term weather and climate change risks, so my focus is not going to be so much on maintaining high fidelity data.

- ☒ I have verified that my primary data source is accessible right now
- ☒ I have reviewed the license and confirmed I can use this data for my project
- ☐ I have examined sample data and understand its structure
- ☒ I have a backup plan if my primary data source becomes unavailable
- ☒ My data volume is sufficient for training a meaningful model
- ☐ I have considered and documented any data quality issues

Backup Data Plan:

ERA5 is the most comprehensive dataset for this kind of task, I would have to refine scope in the event the data does not work. Regardless, there are other climate/weather data sources, and weather models to get data from

✓ Section 6: Technical Approach

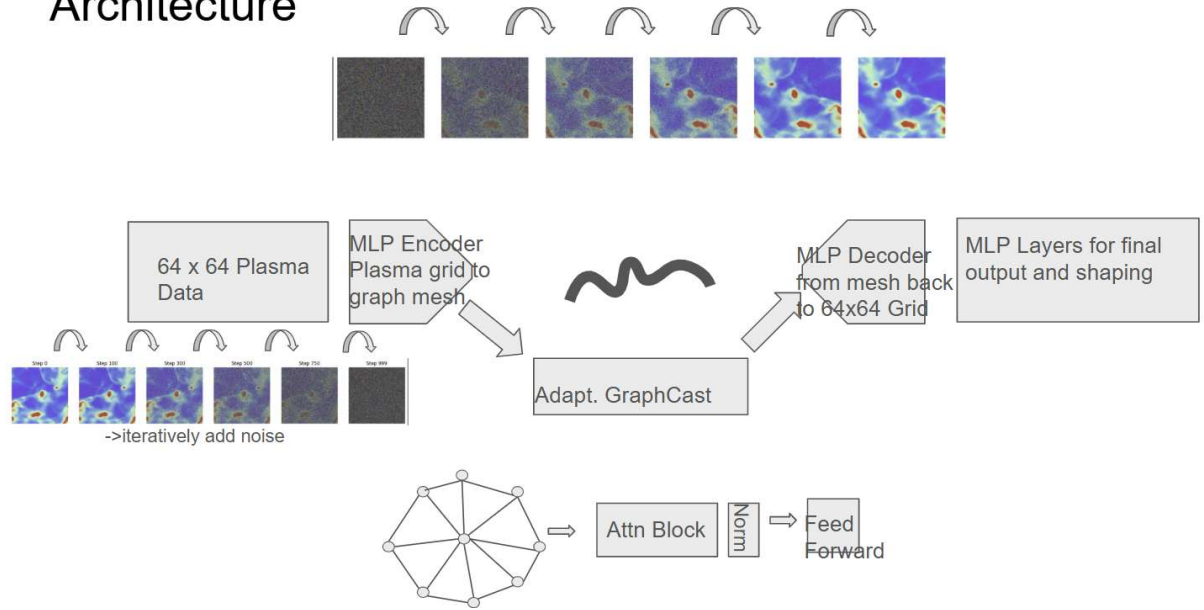
6.1 High-Level Architecture

Describe the overall technical architecture of your system:

1. **Input:** hdf5 data -> Pandas
2. **Processing:** Data transformed into tensor to be encoded to graph
3. **Model:** Gnn/DDPM hybrid
4. **Output:** Numpy array
5. **Interface:** Web interface, API

Architecture Diagram:

Architecture



[Draw or describe your system architecture here]

✓ 6.2 Model Selection

Primary Model Architecture:



Primary : Diffusion - Graph Transformer;

Why: This model has been used successfully by Google in the GenCast model, diffusion models are able to uncover finer visual output/dynamics and handle UQ. The graph architecture enables the discover of relationships between plasma voxels and thus the ability to represent chaotic dynamics. In the future, the graph part will be useful when extending to more complicated geometries (i.e. tokamaks, stellarators, intertial confinement fusion capsules, etc)

Baseline: Numerical sim models, based on partial differential equations and known physics

Alternative 1: Fourier Neural Operator

Why: Instead of representing output, operator models seek to find a representative functional form of the problem, (i.e. surrogate partial diff eqs) that are then used to compute the output. They do this by stacking MLP kernels on top of eachother until the

"universal approximation theorem" criteria is met. NVIDIA used FNOs for weather prediction and select physics problems (i.e. Darcy flow) and achieved great results.

When to try it: If the graph-diffusion model proves too technically difficult or struggles to represent the physics after training, FNO model will be resorted to as it is more well studied for this kind of problem

Alternative 2: Physics Constrained Basic MLP

Why: Simple models that are well studied and easy to debug, just not good at representing more complex physics and are fairly rigid.

When to try it: If scope proves to complex, will reduce data down to 1-D (i.e. density profile) and train simple physics informed MLP to predict

<https://maziarraissi.github.io/PINNs/>

6.3 Training Strategy (if applicable)

Training Approach: from scratch + later fine tuning

Justification:

Problem is niche and I am working with a new model

Hyperparameter Tuning Plan:

Will use Bayes search with W&B platform integrated

6.4 Technology Stack

Complete the following table with your planned tools. You don't have to use the items listed below, they are just provided to give you some ideas for what to include:

Component	Technology	
Programming Language	Python 3.10	Python is a powerful, high-level language well
ML Framework	PyTorch, Numpy	PyTorch is industry standard and well optimiz
Data Processing	Pandas, NumPy, Google Cloud mounting/API	Working with large amounts of numerical, sim
Experiment Tracking	W&B	Reliable framework for model managing and f
Model Registry	W&B Registry	Accessible, organized, collaborative
Version Control	Git + GitHub	Versatile, well-organized platform for storing c
Deployment Platform	Hugging Face	Widely-used for ML models, free hosting
Monitoring	Prometheus	Provides useful physics based and operations
Testing Framework	pytest	Well used testing framework, that can be nice

Component		Technology
Documentation	Quarto	Allows for writing highly-technical, mathematical documents

✓ Section 7: Evaluation Strategy

7.1 Success Metrics

Define **quantitative metrics** for evaluating your project:

Metric	Target Value	Measurement Method	Priority
RMSE Pixel Wise	Error from 64b64 velocity, mag field output to input	Test set eval	Must have
Physics Enforcement	Magnetic field near divergentless $-1e-5$	Test set eval	Must have
Speed	100x speed up compared to traditional solv	test set eval/timing	Should have

7.2 Evaluation Methodology

Train/Validation/Test Split:

60% train, 15% test, 25% eval

Cross-Validation Plan:

Not formally "cross-validated", instead, will take subsets from distinct time windows in simulation data and train on those

Evaluation Frequency:

To be determined based on data intensity and model performance. Tentatively, every 100 batches there will be an eval

7.3 Baseline Comparisons

You must compare your model against baselines. List your planned comparisons:

Baseline	Expected Performance	Purpose
Numerical Solver	Comparable output, faster	Model is physically based
2D-FNO	Comparable, more detailed dynamics	Comparable to more established ML frameworks

7.4 Qualitative Evaluation

How will you evaluate aspects that can't be measured numerically?

The output of the diffusion model should be visually smooth, consistent.

Section 8: Deployment Strategy

At a high level, please detail the way in which you plan to deploy your project. You don't need to know the details of how to do this yet, but it's important to think about the end state of how you would have users interact with your system once it's completed. This can affect many other components of your project including your requirements, system architecture, and timeline. For example:

Deployment Target: Web app for visualization and model interaction. Meant as tool for plasma physicists interested in using ML to cope with experimental data and to serve as fine-tunable, foundational model for physicists

Deployment Platform:

Likely a mixture of cloud platform and simple app frameworks. Depending on the complexity of the model, an inference server, akin to NVIDIA's Triton, may be necessary.

✓ Section 9: Timeline & Milestones

9.1 Project Phases

Break your project into phases aligned with the semester schedule:

Phase	Week(s)	Deliverables
1: Setup & Data	1-3	Set up Git, Data loaded on Cloud and accessible pipeline, Demonstration
2: Baseline Model	3-4	Develop model and ensure it is compatible with the data
3: Model Development	5-7	Train model and fine-tune
4: Iteration & Improvement	8-10	Continue improving model and tune hyper-parameters. Extend to another dataset
5: Deployment & Polish	10-13	Migrating model to Hugging face, develop visualization utensils
6: Documentation & Presentation	14-15	Technical report in Quarto, fully deployed on Hugging face and a web interface

9.2 Weekly Goals

For the first four weeks, define specific goals:

Week 1:

- ☒ Sec 1-5 planning, Literature review

Week 2:

- ☒ Literature Review + Planning

Week 3:

- ☒ Finish rest of planning
- ☐ Set up Google cloud platform for data
- ☐ Mount Google cloud and load data

Week 4:

- ☐ Establish model
- ☐ Pass data through
- ☐ Establish physics constraints

9.3 Checkpoint Submissions

List what you will submit at each checkpoint that you have identified:

Checkpoint	Due Date	Deliverables
Project Proposal (this document)	2/15	This completed notebook
Data & Baseline Review	2/20	Data is accessible through cloud and translated into Numpy framework
Midterm Progress Report	3/10	Model is established and can generate roughly relevant output
Deployment Demo	3/24	Model is trained, visualizable, and already uploaded to hugging face
Final Submission	5/1	Web interface where user can inference from data slices, and see mode

✓ Section 10: Risk Assessment

10.1 Risk Identification

Identify potential risks and their mitigation strategies:

Risk	Likelihood (L/M/H)	Impact (L/M/H)	Mitigation Strategy
Technical skill	M	H	Intense literature rev, taking from established moc
Data complexity	M	H	Transform data through different means to simplif
Model underperformance	M	H	Meeting all other benchmarks ahead of time to fo
Physics representation	M	M	Use loss functions from other plasma physics info

10.2 Assumptions

List assumptions you are making that, if wrong, could affect your project:

1. Diffusion models work well for physics
2. The model is trainable with reasonable resources
3. Data is physically accurate, consistent

4. GNNs are integratable with diffusion models

10.3 Dependencies

List external dependencies that could affect your timeline:

Dependency	Type	Criticality	Alternative
Google TPUs will remain available	Platform	Low	Use school supercomputer resources
Google cloud data storage affordable	Data storage	Low	Obtain hard drive to store data
Python itself is sufficient for this design	Platform	Medium	Spend time working under the hood with C to op

Appendix A: Project Complexity Guidelines

Your project should fall within the "Appropriate" range:

Level	Characteristics	Example
Too Simple	Tutorial-level, no novel combination, single notebook	Image classifier with dataset from Kaggle
Appropriate	Real-world problem, multiple components, deployed	Plant disease detector with API, monitoring, and